

# Manishika Balamurugan

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## Education

### University of Michigan - Ann Arbor

Expected May 2027

Bachelor of Science in Computer Science & Biopsychology, Cognition, and Neuroscience. (GPA: 3.91/4.00)

Ann Arbor, MI

- **Honors:** Rogel Scholar (Full Ride), William J. Branstrom Prize (Top 5%), University Honors

## Experience

### Sartorius

Jan 2026 – May 2026, Aug 2026 – Dec 2026

Student Software Engineer, UM CoE Multidisciplinary Design Program

Ann Arbor, MI

- Designing containerized backend systems (Docker, Node.js) for remote lab monitoring and control.
- Building secure REST APIs and a React dashboard for real-time telemetry and configuration
- Collaborating with industry engineers to align architecture with commercial products.

### American Airlines Sponsored Project

Jan 2026 – Present

Student Machine Learning Engineer, Michigan Data Science Team

Ann Arbor, MI

- Developing a predictive system to minimize high-cost flight simulator cancellations using historical logs.
- Building a hybrid ML pipeline integrating Random Forests with a GenAI layer to extract features from unstructured pilot notes.
- Partnering with stakeholders to deploy production-ready models for optimized training scheduling.

### Michigan Daily

Sep 2025 – present

Web Developer

Ann Arbor, MI

- Redesigned frontend gameplay components for Ann Arbor Geuguessr game using Svelte and TypeScript.
- Built a SQL-backed daily game generation system to be used for publication.

## Projects

### Stock Market Matching Engine | C++, STL, Priority Queues

- Built a C++ matching engine to process limit orders in  $O(\log n)$  time using price-time priority logic.
- Implemented four custom priority queue architectures to minimize execution latency and memory overhead.
- Developed features to track market medians and "Time Traveler" profit opportunities within historical trade data.

### Replica of GPT-2 | PyTorch

- Implemented a GPT-2-style Transformer from scratch, including multi-head self-attention and positional encodings.
- Built a custom training loop, optimizer, and evaluation pipeline.

### Brain Tumor Detection & Classification | PyTorch

- Achieved 99.7% binary and 95.7% multi-class accuracy with CNNs (ResNet34, ConvNeXt).
- Applied transfer learning, augmentation, and evaluation metrics.

## Research

### Computational & Cognitive Neuroscience Lab, University of Michigan

Aug 2024 – Present

Research Assistant

Ann Arbor, MI

- Built and maintained R-based data processing pipelines for MRI datasets, ensuring reproducibility and accuracy.
- Conduct preprocessing, quality control, and statistical analysis for hundreds of samples.
- Trained an internal AI assistant to streamline literature review and lab communication workflows.

### Regenerative Imaging Laboratory, University of Pittsburgh

May 2025 – Aug 2025

Research Fellow

Pittsburgh, PA

- Analyzed hippocampal connectivity in 18 brains using diffusion MRI and tractography software.
- Wrote MATLAB scripts for clustering neural connections and modeling network patterns.

## Technical Skills

**Languages:** Python, C++, JavaScript, TypeScript, SQL, C, R

**Frameworks/Libraries:** React, Flask, Node.js, FastAPI, PyTorch, Svelte, scikit-learn, OpenCV

**Tools:** Docker, Git, Jupyter, REST APIs, MATLAB

**ML/DS:** CNNs, Transformers, Transfer Learning, Data Pipelines